Python Assignment
Topic-7: Functions and Methods

Q1. Defining and calling functions in Python.

Ans.

Functions are blocks of code used to perform a specific task. They help in dividing a large program into smaller parts. In Python, functions are created using the def keyword. After defining a function, we can call it whenever needed. Functions reduce code repetition and make programs easy to manage. They also improve readability and make debugging easier. A function can also return a value after execution. Using functions makes the program organized and reusable.

Example:

```
def greet():
   print("Hello Anshu")

greet()
```

Q2. Function arguments (positional, keyword, default).

Ans.

Arguments are values passed to a function when it is called. Positional arguments are passed in the correct order of parameters. Keyword arguments are passed using names, so order does not matter. Default arguments have a predefined value which is used if no value is given. Arguments make functions flexible and useful. They allow the same function to work with different values. Understanding arguments is important for writing better programs.

Example:

```
def add(a, b=10):
   print(a + b)

add(5)
add(5, 3)
```

Q3. Scope of variables in Python.

Ans.

Scope means the area where a variable can be used. There are two main types of scope: local and global. Local variables are created inside a function and can be used only within that function. Global variables are created outside functions and can be used anywhere in the program. Proper use of scope helps in avoiding errors. It also helps in writing secure and clean code. Understanding scope is important in large programs.

Example:

```
x = 20

def show():
   y = 10
   print(x, y)

show()
```

Q4. Built-in methods for strings, lists, etc.

Ans.

Python provides many built-in methods to work with data types like strings and lists. These methods help in modifying, searching, and processing data easily. For example, string methods can convert text to uppercase or lowercase. List methods can add or remove elements. These methods save time and reduce code length. They are very useful in everyday programming tasks. Learning these methods helps in writing efficient programs.

Example:

```
text = "python"
print(text.upper())

nums = [1,2]
nums.append(3)
print(nums)
```